The Robert M. Buchan Department of Mining

Queen's University

MNTC 313 – Assignment 12

Falcon 9 is a two-stage launch vehicle designed and manufactured by SpaceX. It is partially reusable, with the first stage capable of re-entering the atmosphere and landing vertically after separating from the second stage.

https://www.youtube.com/watch?v=DLVsvXVPIrQ

The first stage is coming back and entering the troposphere with a pure vertical velocity of **1500 km/h**, at a height of **10 km**. The stage presses on the gas, which applies **174850 N** of constant force on the vehicle, directed up. The stage stays on the gas for a total of **60 seconds** non-stop. You should indicate at which height the stage starts to press on the gas (for controlleddescent). It is assumed that the vehicle maintains pure vertical velocity, without considering horizontal movement during the landing. The mass of the first stage is **15600 kg.**

Using MATLAB, write a program that simulates the motion of the first stage re-entering the troposphere. **Your program should run while the vehicle is in motion and stop once the vehicle comes to the landing platform at sea level (0 km) [**be careful since it may happen that the vehicle never reach 0 level by a coding error, and the ***while*** never ends, if you forget to stop the propelling**]. A starter code has been provided on OnQ, which provides user input statements for the height at which the controlled-descent starts, and the fluid density. Assume fluid frictional force.** For large objects moving through air, the air resistance is approximately proportional to the square of the velocity. The form of the resistance is

$$f_{\text{drag}} = -\frac{1}{2} C\rho A v^2$$

Where ρ is the fluid density, A the cross-sectional area, and C is a numerical drag coefficient. The drag coefficient C is 0.5 for a spherical object and can reach 2 for irregularly shaped. We

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assume a drag coefficient **C of 1.0** for the first stage. **The negative sign implies that it is always directly opposite the velocity. Consider a cylindrical shape with diameter of 3.7 m.** Your simulation should store the following values on each iteration: **acceleration, height distance, velocity, and time**. Conduct simulations with a dt of **1 s, 0.1 s and 0.01 s**. Use **g = 9.8 m/s^2.** Your program should output the following:

- The **time step, total time and final position** for each time step used
- The **landing speed** reached by the vehicle during its motion down to the platform, in **km/h** [only for the **0.01 s** case]
- If the speed limit for a safety landing of the vehicle is **10 km/h**, indicate whether it was exceeded. Print **Non-successful landing** if the landing speed limit is exceeded**.** Print **Successful landing** if the speed limit is obeyed. [only for the **0.01 s** case]

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An example of your output should look like:

```
Enter controlled-descent height (m):2500

Enter fluid density (kg/m^3):1.21

dt = 1.00 s, Total time = 80 s, Position = -11 m dt
= 0.10 s, Total time = 105 s, Position = -0 m dt =
0.01 s, Total time = 88 s, Position = -0 m
landing speed: 5 km/h
Successful landing
```